SIMATS ENGINEERING

Department of Computer Science and Engineering

CSA15 Cloud Computing and Big Data Analytics

CO2 AT3 - Capstone Cloud Infrastructure Case Analysis and Defense

Editable student template for application-based cloud virtualization and SDDC/cloud operations defense

Assessment Metadata

Course Code	CSA15
Course Title	Cloud Computing and Big Data Analytics
Assessment	CO2 AT3 - Capstone Cloud Infrastructure Case Analysis and Defense
Submission Type	Individual digital case analysis and infrastructure defense based on the same capstone title used in CO1, CO2 AT1 and CO2 AT2
Faculty	Dr C. Rajesh Babu
Employee ID	SSETSCS415
Submission Mode	Completed DOCX template and GitHub repository link through Google Classroom

How This Template Works

- Use the same capstone title used in CO1, CO2 AT1 and CO2 AT2.
- Do not recalculate all VM sizing or repeat all configuration screenshots. Summarize the final AT1/AT2 evidence and use it for design defense.
- Answer all five questions with application-oriented reasoning connected to the capstone product.
- Include sustainable cloud operation decisions wherever relevant: right-sized resources, log retention, storage lifecycle, scheduled shutdown, API-call reduction and VM/container/serverless comparison.
- Paste the final topology diagram in the given response area. The sample diagram is only a structure; redraw it for the student's own product.
- Upload this completed document and supporting evidence to GitHub, then submit the repository link through Google Classroom.

Assessment Focus
AT1 answered what VM is required. AT2 showed configuration or simulation evidence. AT3 asks whether the student can defend the complete infrastructure design under application, security, operations and failure conditions.

Student Name	Blessy S
Register Number	192421053
Class / Slot	Slot B

Student and Capstone Details

Capstone Title	Cloud security
CO1 Evidence Link	https://github.com/192421053simats-cmd/Cloud-computing-
CO2 AT1 VM Recommendation	vCPU: ____ RAM: ____ GB Storage: ____ GB VM Class: ____
CO2 AT2 Evidence Link	https://github.com/192421053simats-cmd/Cloud-computing-
GitHub Repository Link	https://github.com/192421053simats-cmd/Cloud-computing-
Submission Date	13.08.26

Evidence Carry-Forward Summary

Evidence Source	Student Entry	How It Supports AT3 Defense
CO1 concept map / presentation	Project concepts	Shows understanding
CO2 AT1 sizing workbook	Resource calculations	Justifies resource selection
VM calibration sheet	VM configuration	Proves proper setup
CO2 AT2 practical evidence	Implementation & testing	Proves practical work
Capstone architecture decision	Final architecture	Justifies design choice

Q1: Virtualization Value and Capstone Cloud Deployment Summary

Explain your capstone product, expected workload, cloud need and deployment context. Analyze how virtualization improves cost efficiency, resource utilization, workload isolation, infrastructure flexibility and cloud deployment readiness for your product.

Design Point	Student Response
Capstone product and users	Cloud Security system for protecting cloud resources and users.
Expected workload	User authentication, security monitoring, threat detection

	and data protection.
Cloud need	Cloud provides scalable, secure and flexible infrastructure for the system.
Virtualization value	Virtual machines provide isolated and controlled environments for security testing.
Cost and utilization benefit	Reduces hardware cost and improves resource utilization.
Isolation and flexibility benefit	Provides workload isolation and allows easy scaling and configuration.

Case Analysis Response: Write 8-12 technical lines connecting virtualization value to your capstone product.

1. Virtualization allows the Cloud Security system to run in an isolated virtual environment.
2. Virtual machines help protect workloads from unauthorized access.
3. Resources such as CPU, RAM, and storage can be allocated efficiently.
4. Virtualization reduces the need for dedicated physical hardware.
5. Multiple security workloads can run on separate virtual machines.
6. VM isolation limits the impact of security threats or failures.
7. The cloud environment can be scaled easily when users increase.
8. Virtual machines can be configured and tested without affecting other systems.
9. This improves flexibility, resource utilization, and security testing.
10. Therefore, virtualization supports a cost-effective and secure Cloud Security deployment.

Q2: Rapid Provisioning, Testing and Virtual Machine Lifecycle Case

Using your CO2 AT1 VM recommendation and CO2 AT2 practical evidence, explain how virtual machines, VM images, templates, snapshots, clones or containers support rapid provisioning, testing, deployment, rollback and maintenance for your capstone product.

Lifecycle Element	AT1/AT2 Evidence Used	Capstone Use / Justification
VM image or base environment	Ubuntu VM setup	Provides a secure base environment.
Template or repeatable setup	VM configuration	Enables quick and consistent deployment.
Snapshot / rollback	VM snapshot	Allows recovery after security testing
Clone / testing copy	Cloned VM	Enables safe testing without affecting the main VM.
Container or VM decision	VM-based deployment	Provides strong isolation for security
Maintenance and update plan	VM updates and security patches	Keeps the system secure and updated.

Case Analysis Response: Explain how lifecycle practices reduce deployment delay and risk.

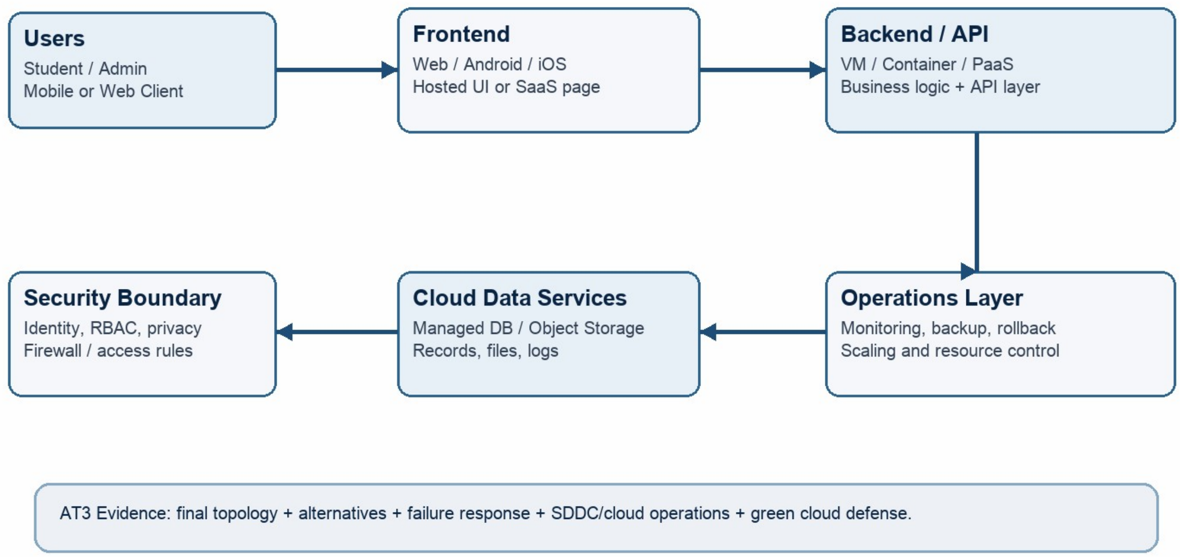
VM lifecycle practices make Cloud Security deployment faster and safer. VM images provide a ready-to-use secure environment. Templates allow consistent and rapid VM deployment. Snapshots provide quick rollback after failed updates or security tests. Cloning creates a separate environment for testing. VM isolation protects the main system from testing risks. Regular updates and patches reduce security vulnerabilities. These practices reduce deployment delay, configuration errors, and operational risk.

Q3: Final Cloud Deployment Topology and SOA/API Service Integration Case

Draw and explain the final deployment topology for your capstone product. Show frontend, backend/API, database, storage, authentication, notification, analytics and any VM/container/managed cloud service components. Explain how SOA/API-based integration connects these modules and supports modular cloud service design.

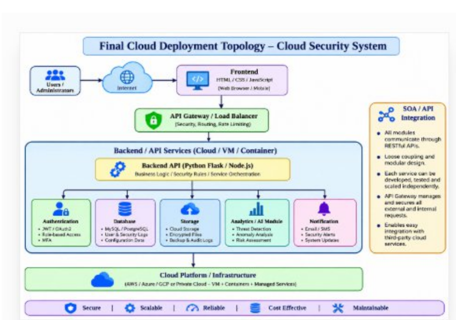
Capstone Cloud Deployment Topology - Sample Structure

Students should redraw this with their own modules, services, VM/container choices and security boundaries.



Architecture Component	Technology / Service	VM / Container / Managed Service / SaaS	Integration Method
Frontend	HTML, CSS, JavaScript	VM / Web Server	REST API
Backend/API	Python Flask	VM / Container	REST API
Database	MySQL	Managed Service	API / SQL
Storage	Cloud Storage	Managed Service	API
Authentication	JWT Authentication	Backend Service	REST API
Notification	Email Service	SaaS	API
Analytics / AI module	Python Security Analytics	Container	REST API

Paste or draw the final capstone deployment topology diagram here.



SOA/API Integration Explanation: Explain how services communicate and why modular service design is useful.

SOA/API allows different cloud services to communicate through REST APIs. The frontend sends requests to the backend API, which connects to authentication, database, storage, analytics, and notification services. Modular design allows each service to be developed, tested, updated, and scaled independently. This improves **security, flexibility, scalability, and maintenance** while reducing system complexity..

Q4: Design Alternatives, Identity, Access and Federated Cloud Privacy Case

Compare at least two deployment alternatives for your capstone product, such as single VM, multi-tier VM, containerized deployment, managed backend, federated cloud or hybrid cloud model. Justify the selected design by analyzing identity management, role-based access control, federated access, data sharing, privacy protection and security risks.

Deployment Alternative	Advantages	Limitations / Risks	Identity and Privacy Impact
Alternative 1	Simple and low cost	Limited scalability	Basic access control
Alternative 2	Flexible and scalable	More configuration required	Better isolation and access control
Selected design	Scalable, flexible and secure	Requires monitoring	Strong authentication, RBAC

Security Area	Student Decision
User roles	Admin and normal user
Authentication method	Username/password + JWT
Role-based access control	Access based on user role
Federated access / partner access	OAuth 2.0 can be used
Sensitive data handled	User credentials and security logs
Privacy protection decision	Encryption and restricted access

Case Analysis Response: Defend why the selected design is better for identity, access and privacy.

The selected containerized design provides better isolation and scalability. JWT authentication verifies users, while RBAC restricts access based on roles. Sensitive data is encrypted and access is limited to authorized users. This reduces identity, privacy and security risks.

Q5: Software-Defined Datacenter and Cloud Operations Defense Case

Analyze how your infrastructure will respond to operational and failure scenarios such as VM failure, database growth, traffic spike, credential leakage, service outage, resource contention or cost increase. Explain how SDDC concepts and cloud operations practices such as automated provisioning, software-defined networking, software-defined storage, monitoring, logging, backup, rollback, scaling, resource control and maintenance improve the final infrastructure design. Include Green Computing decisions such as resource right-sizing, storage lifecycle control, log retention, API-call reduction, scheduled shutdown, VM/container/serverless comparison and carbon/energy proxy indicators where relevant.

Operational / Failure Scenario	Expected Impact	SDDC / Cloud Operations Response	Evidence or Control
VM failure	Service downtime	Backup and recovery	VM backup
Database growth	Storage increase	Secure storage scaling	Encryption
Traffic spike	Slow service	Auto-scaling + firewall	Security rules
Credential leakage	Unauthorized access	Key rotation + RBAC	IAM settings
Service outage	Service unavailable	Monitoring + failover	Monitoring logs
Resource contention	Poor performance	Monitoring + failover	VM limits
Cost increase	Higher cost	Monitor and shut down unused	Cost reports

Operational Readiness Area	Student Plan
Automated provisioning	Automate secure VM deployment
Software-defined networking	Configure firewall and secure network rules
Software-defined storage	Use encrypted cloud storage
Monitoring and logging	Monitor security logs and activities
Backup and rollback	Maintain backups for recovery
Scaling and elasticity	Scale resources securely when needed
Resource control and maintenance	Apply access control and regular updates
Green Computing and sustainability control	Reduce unnecessary cloud resource usage

Green Cloud Decision	Student Justification
VM right-sizing from CO2 AT1	Use only required resources
VM vs container vs serverless choice	Choose containers for efficiency
Scheduled shutdown / idle-resource control	Shut down idle VMs
Log and file retention policy	Delete old unnecessary logs
Storage lifecycle and backup cleanup	Remove old backups
API-call or polling reduction	Reduce unnecessary API calls
Carbon/energy proxy indicator to track	Monitor CPU and energy usage

Final Infrastructure Defense Statement: State whether the design is ready for prototype, pilot or production. Defend reliability, security, SDDC/cloud operations and Green Computing improvements required next.

Our **Cloud Security project is ready for the prototype stage**. The design provides basic reliability through backups and recovery. Security is improved using **IAM, RBAC, encryption and firewall rules**. SDDC operations support monitoring, resource control and automation. Green Computing can reduce energy use by shutting down unused resources. For a pilot, we need stronger monitoring, auto-scaling and disaster recovery. For production, we need continuous security testing, high availability and cost

CO2 Reflection and Individual Defense

This section must be individually written. It should show what the student understands from the complete CO2 chain: infrastructure sizing, practical configuration evidence and final operations defense.

Reflection Prompt	Student Response
What is the strongest infrastructure decision in your capstone design?	Secure cloud VM with proper access control.
Which CO2 concept became clearer after AT1, AT2 and AT3?	Right-sizing and resource optimization.
Which risk is most serious for your capstone: failure, cost, privacy, scaling or sustainability?	Privacy and security.
What evidence would you show during viva to defend your infrastructure design?	VM settings, firewall rules and security logs.
What will you improve before final capstone deployment?	Improve monitoring, backup and security.

Student-Facing Assessment Criteria

Criteria	Descriptor
Capstone continuity	The answer uses the same capstone title and connects CO1, CO2 AT1 and CO2 AT2 evidence.
Virtualization and lifecycle reasoning	VM, image, snapshot, clone, container or managed-service decisions are technically justified.
Topology and integration clarity	Architecture diagram and SOA/API explanation show clear module interaction.
Identity, access and privacy	Roles, authentication, RBAC, federated access, sensitive data and privacy decisions are defended.
SDDC/cloud operations	Failure response, monitoring, logging, scaling, backup, rollback and resource control are addressed.

Green Computing defense	Right-sizing, retention, API efficiency, idle-resource control and sustainable cloud choices are explained.
Submission quality	GitHub, template completion, evidence references and individual reflection are complete.

GitHub and Google Classroom Submission

Repository Item	Expected Content
README.md	Capstone title, student details, final VM recommendation, deployment topology summary and AT3 evidence index.
Completed AT3 document	This completed editable DOCX or exported PDF if instructed.
Diagram image	Final topology diagram if created separately.
Evidence references	Links or screenshots from CO1, CO2 AT1 and CO2 AT2 as needed.
Green Computing note	Short README section explaining sustainable cloud choices and resourcecontrol decisions.
Google Classroom	Submit the GitHub repository link through GCR.

Student Assurance

I confirm that this case analysis, design reasoning, topology diagram and final defense statement were prepared individually by me for the stated capstone title. I have not uploaded passwords, tokens, private keys or sensitive account information.

Student Name	Blessy S
Register Number	192421053
Signature	**blessy**
Date	13.08.26

Faculty Verification

Faculty Name	Dr C. Rajesh Babu
Employee ID	SSETSCS415
Constructive Feedback / Remarks	
Strength Observed	
Improvement Suggested	
Signature / Date	